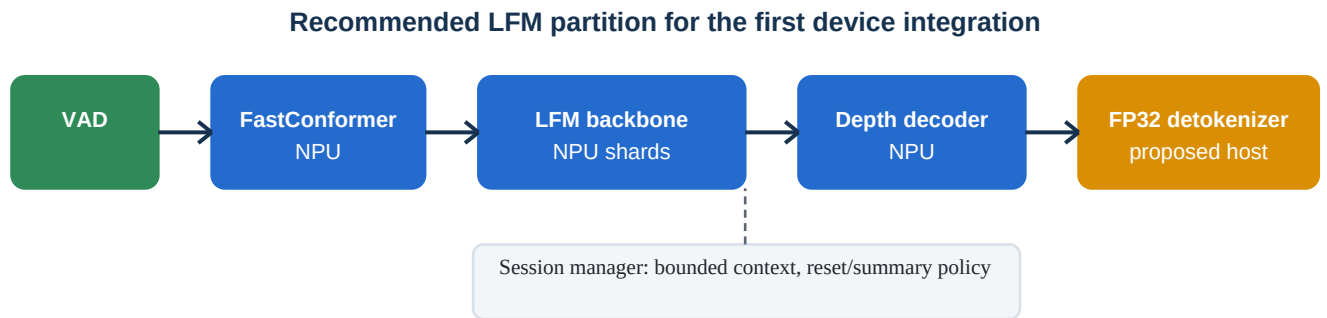


LFM2.5-Audio on AI Glasses

Feasibility, Quality, and Baseline Decision



Public technical report

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Decision: continue LFM as the integrated research baseline, while profiling a sub-1B modular pipeline as deployment control and demo insurance.

Hardware boundary: Qualcomm AI Hub exposed QCS8550 (Proxy), Android 12, Hexagon v73. No AR1/AR1+/5100 target was available. All proxy figures are placement evidence, not AR1 performance.

1. Executive decision

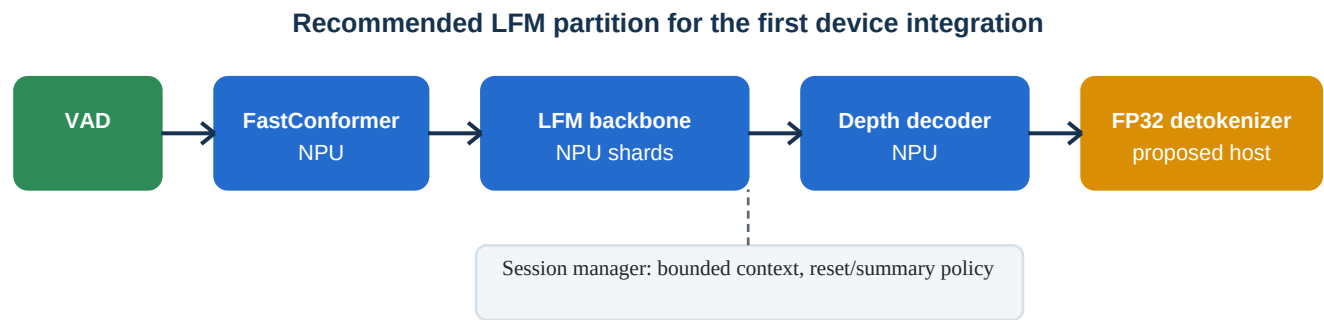
Proceed with LFM2.5-Audio as the integrated research baseline, but do not call it glasses-ready yet. All tested fixed-shape neural partitions compiled to strict NPU contexts with zero CPU fallback on the QCS8550 proxy. The full official BF16 model also completed ASR and persistent two-turn speech interaction on NVIDIA L4.

The main blockers are target-hardware verification, target-QNN resident memory, a production cache interface, and the output-audio path. The FP16 NPU detokenizer fails numerically despite full placement. The first device integration should keep the official FP32 detokenizer outside the FP16 HTP path until a host implementation is measured and accepted.

In parallel, profile Moonshine Base -> Qwen3-0.6B -> Pocket TTS. It is the strongest weight-footprint challenger. Keep Qualcomm-native Zipformer and Piper TTS as conservative swaps if export or host integration becomes the schedule risk.

2. Confirmed usage mode

The product mode is VAD-gated activation followed by a continuous conversational session. This avoids constant full-model inference while idle, but it does not bound active-session context. The session manager needs an explicit reset, summarization, or eviction policy.



The proposed host detokenizer is an architectural recommendation, not a measured CPU-host result on the glasses.

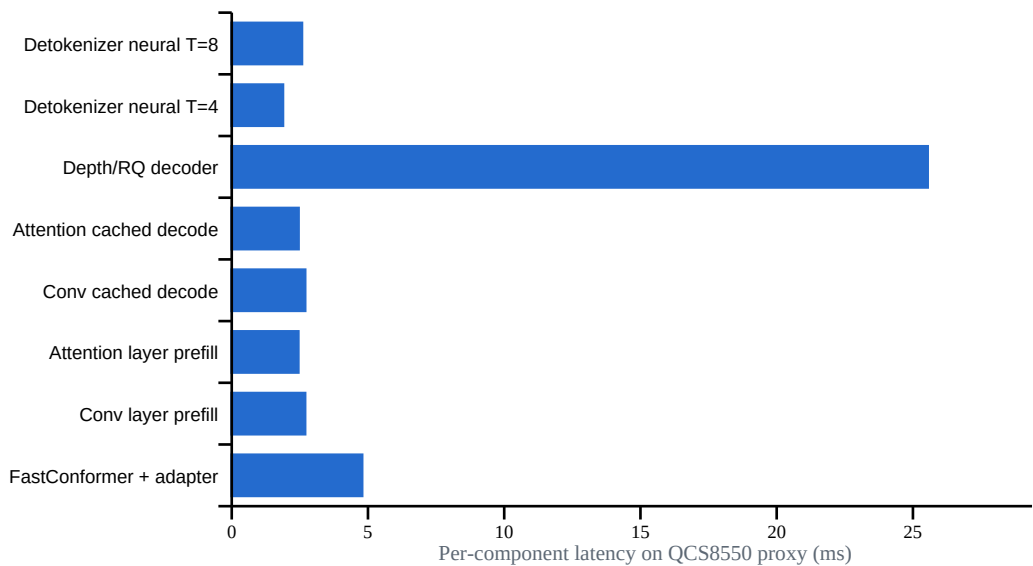
3. Experiment scope

Evidence class	What is included
Local	Eight official-weight fixed-shape PT2 exports; repeated Q4_0/KV-cache matrix on Apple M2 Metal; static memory ledger.
Qualcomm proxy	Strict NPU placement, component-only latency/memory, and local-vs-remote numerical comparison.
Reference GPU	Full BF16 ASR plus two-turn interleaved text/audio generation on one NVIDIA L4.
Quality	Downstream encoder sensitivity, exact depth tokens, generated-wave integrity, and a small matched ASR diagnostic.
Not measured	Full AR1 pipeline, host/NPU transfers, real glasses RSS, power, thermals, VAD, microphone-to-speaker latency.

4. Qualcomm proxy component results

Every tested strict graph placed fully on the Hexagon NPU of QCS8550 (Proxy). Placement and numerical acceptance are separate: the detokenizer demonstrates that a graph can be fully placed and still be unusable.

Component	NPU/CPU	Latency	Peak	Numerics
FastConformer + adapter	1215/0	4.786 ms	87.0 MiB	Mismatch
Conv layer prefill	36/0	2.693 ms	92.2 MiB	Pass
Attention layer prefill	62/0	2.444 ms	91.5 MiB	Pass
Conv cached decode	32/0	2.695 ms	91.9 MiB	Pass
Attention cached decode	61/0	2.452 ms	87.0 MiB	Pass
Depth/RQ decoder	2953/0	25.540 ms	92.0 MiB	Pass
Detokenizer neural T=4	371/0	1.880 ms	88.0 MiB	Mismatch
Detokenizer neural T=8	371/0	2.577 ms	87.4 MiB	Mismatch



Do not add these latency or peak-memory values into a full-system estimate. The components have different shapes, repeated invocation counts, compiled-context overheads, and transfer boundaries.

5. Functional full-model golden

Metric	Turn 1	Turn 2
First text	97.97 ms	56.49 ms
First audio token	339.67 ms	225.87 ms
Generated steps/s	15.74	16.35
Decoded waveform	11.84 s	8.96 s

The full model used 3.034 GB steady CUDA allocation and 6.216 GB peak allocation. These are reference-GPU allocator figures, not mobile RSS. The official output detokenizer remained FP32 on the L4 GPU.

6. Quality findings

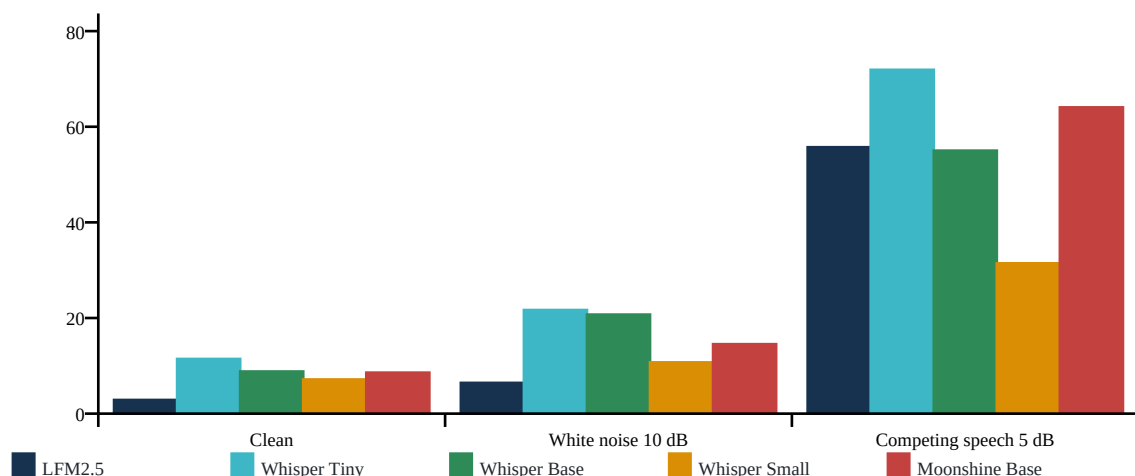
FastConformer proxy features: cosine 0.998424, NRMSE 0.06976, and 100% nearest-frame temporal identity over the ten captured frames. A frozen downstream LFM sensitivity test preserves 16/17 top-1 decisions. The only branch change is 'seems' to 'looks'; the golden choice stays in the remote top five for every step.

Interpretation: the encoder path is promising enough for the baseline, but exact tensor closeness is not the final gate. Every quantization change must be checked on transcripts, tokens, and task-level behavior.

The depth/RQ decoder returns exact audio code tokens. The FP16 NPU detokenizer remains unusable on real generated codes: waveform cosine 0.0052, NRMSE 1.010, and SI-SDR -45.74 dB for the first eight valid turn-1 frames. The graph still places 371/371 runtime layers on NPU at 2.564 ms, proving that placement alone is insufficient.

Small matched ASR diagnostic

The same 18 LibriSpeech dummy utterances were tested clean, with deterministic 10 dB white noise, and with 5 dB competing speech. WER is normalized and lower is better. This is a smoke diagnostic, not a product benchmark.



WER on the same 18-utterance diagnostic; lower is better

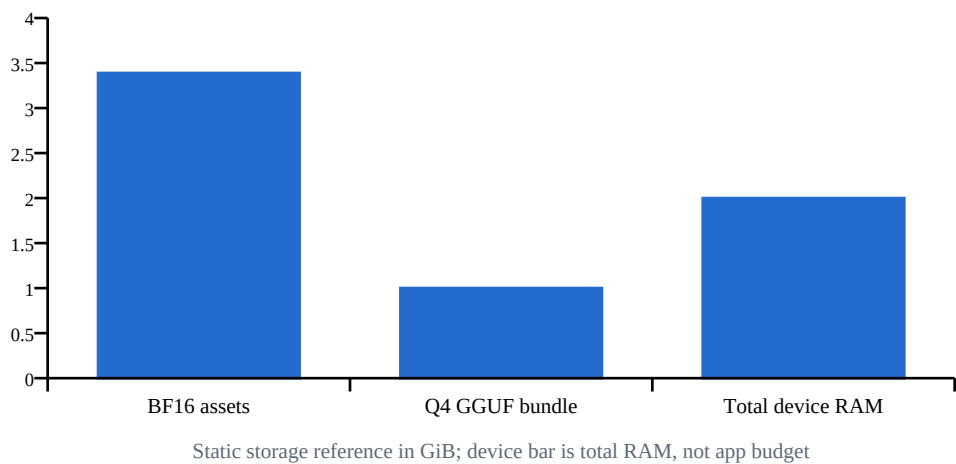
Model	Clean	White 10 dB	Competing 5 dB
LFM2.5	2.86%	6.43%	55.71%
Whisper Tiny	11.43%	21.67%	71.90%
Whisper Base	8.81%	20.71%	55.00%
Whisper Small	7.14%	10.71%	31.43%
Moonshine Base	8.57%	14.52%	64.05%

On this small diagnostic, LFM is strongest on clean speech and 10 dB white noise, while Whisper Small is clearly strongest under competing speech. Moonshine remains the footprint challenger, not a demonstrated multi-speaker quality winner.

Generated-audio intelligibility proxy

Whisper Small retranscription gives 0.00% WER for turn 1 and 21.43% WER for turn 2 against LFM's own generated text. The WAVs are finite, 24 kHz mono, and unclipped. This is an audio-text consistency proxy, not a naturalness or MOS score.

7. Memory and context budget



The pinned BF16 runtime checkpoint set occupies 3.390 GiB in static files. The complete local Q4 GGUF bundle occupies 1.001 GiB, a 3.39x smaller package-to-package comparison across different formats and backends. Static file size is not process RSS; the BF16 assets cannot all be resident inside a 2 GiB RAM budget.

Context positions	Ideal FP16 payload	Ideal INT8 payload
4,096	48 MiB	24 MiB
8,192	96 MiB	48 MiB
16,384	192 MiB	96 MiB
32,768	384 MiB	192 MiB

The fixed convolution cache is only about 0.12 MiB at BF16. Active-session growth is dominated by the six attention-layer KV caches. These are tensor-payload estimates; actual quantized runtime buffers can differ because of scales and alignment. VAD reduces idle work but does not remove this requirement.

8. Local Q4 inference-technique matrix

The official Q4_0 GGUF bundle was run 40 times across two official audio samples with FP16, Q8_0, and Q4_0 KV caches and a flash-attention control. 40/40 normalized transcripts match exactly. This is an Apple M2 Metal sanity check, not Qualcomm QNN/AR1 evidence or a general quality benchmark.

KV	Flash	Sample	Exact	KV MiB	Encode ms med/p95	Gen tok/s med/p95	Model ms med/p95
FP16	off	asr	5/5	48.0	449.0 / 629.8	24.3 / 25.6	3425.8 / 3555.5
FP16	off	question	5/5	48.0	140.0 / 298.4	29.1 / 57.4	816.9 / 1590.3
FP16	on	asr	5/5	48.0	439.0 / 592.4	23.2 / 26.4	3389.3 / 3855.5
FP16	on	question	5/5	48.0	172.0 / 196.8	28.6 / 38.8	891.4 / 1096.6
Q4_0	on	asr	5/5	13.5	413.0 / 476.8	25.7 / 26.1	3122.0 / 3285.7
Q4_0	on	question	5/5	13.5	139.0 / 148.8	32.6 / 48.6	738.1 / 904.6
Q8_0	on	asr	5/5	25.5	486.0 / 780.4	26.5 / 28.1	3254.8 / 3780.5
Q8_0	on	question	5/5	25.5	137.0 / 195.4	41.7 / 67.7	627.5 / 982.5

Values are median/p95 over five runs per sample and configuration. CLI model time excludes model load; wall time in the public data includes process startup and file-cache effects. The runner reports Metal fallback hotspots for CONV_2D_DW, ROLL, and UNARY in the audio encoder. Those are not QNN CPU-fallback evidence. No quantized LFM QNN graph has yet been profiled on QCS8550 or the glasses.

9. Additional Qualcomm speech profiles

Model	Encoder/main	Recurrent	Use
Zipformer	8.895 ms / 0.71 s chunk	0.075 ms decoder; 0.187 ms joiner	Streaming fallback
Whisper Tiny	25.307 ms	2.459 ms decoder	Small control
Whisper Base	47.978 ms	4.202 ms decoder	Middle control
Whisper Small FP16	130.318 ms	12.074 ms decoder	Overlap-robust control
Whisper Small W8A16	376.895 ms	7.856 ms decoder	Memory tradeoff
Piper TTS	30.344 ms encoder; 15.189 ms flow	3.018 ms decoder	Audio fallback

All rows are full-NPU QCS8550 proxy scorecards from qai-hub-models 0.57.3. They are fixed component invocations, not end-to-end pipeline latency. Whisper Small W8A16 makes the encoder 2.89x slower while making the decoder about 35% faster, so quantization must be profiled stage by stage.

10. Candidate decision

Candidate	Role	Footprint view	Decision
LFM2.5-Audio	Integrated research baseline	Q4 bundle 1.001 GiB locally	Continue with gates
Moonshine + Qwen3-0.6B + Pocket	Best footprint challenger	Estimated 0.55-0.66 GB weights	Profile next
Zipformer + Qwen3-0.6B + Piper	Conservative toolchain fallback	About 0.84 GB from component cards	Demo insurance
Mini-Omni	End-to-end control	About 976M total estimated	Later control
LLaMA-Omni2-0.5B	Misleading size label	Checkpoint alone 3.857 GB BF16	Exclude

Moonshine/Qwen/Pocket is the recommended next challenger because it creates the most headroom while keeping the text model on a known Qualcomm path. It still needs export, quality, and host-TTS measurements before it can replace LFM.

11. Exact next steps

1. Confirm the physical glasses SoC, app-available RAM, QNN/Voice AI SDK version, and deployment interface.
2. Profile Qwen3-0.6B first, then Moonshine Base encoder/cached decoder, then Pocket TTS on the official host path. Keep Zipformer/Piper ready as swaps.
3. Export four-layer LFM backbone shards and a production cached decoder with explicit position/mask inputs and bounded context.
4. Convert and profile target-QNN quantized encoder, backbone, depth decoder, and caches separately. Run downstream transcript/token checks after every precision change.
5. Measure the official FP32 detokenizer on the intended CPU/DSP path. Do not move it to FP16 HTP until real-code audio fidelity passes.
6. Integrate VAD and measure speech-end to first audible PCM, process RSS, 30-minute temperature, and battery drop on the real glasses.

12. Demo acceptance gates

Gate	Provisional threshold
Memory	Provisional model-process peak $\leq$ 1.3 GB; revise after app RAM is confirmed.
Placement	No unsupported main-backbone op; no undeclared CPU fallback in NPU shards.
ASR quality	Quantized WER increase $\leq$ 0.3 percentage points on each evaluation split.
Responsiveness	First audible PCM $\leq$ 500 ms after VAD close on the actual device.
Context	Two-turn retention no more than five points below the BF16 golden set.
Audio	No NaN/Inf, truncation, or material audio-text consistency regression.

13. Limitations and release notes

- No exact AR1/AR1+/5100 target was available; QCS8550 proxy latency is not AR1 latency.
- Component latency and memory cannot be summed into an end-to-end figure.
- Backbone evidence is representative rather than one full compiled network.
- The small ASR diagnostic uses synthetic noise and is not a glasses microphone benchmark.
- Host FP32 detokenizer viability remains proposed rather than measured on target.
- The Q4 matrix uses two official samples on Apple M2 Metal; it does not establish general quality, Qualcomm placement, or glasses memory.

- LFM Open License v1.0 contains a commercial-use revenue threshold. Candidate components have separate attribution and license terms.

14. Reproducibility

Item	Pinned value
LFM model revision	c362a0625dfe45aa588dce5f0ada28a7e5707628
Q4 GGUF / runner	7d525f883a077e20afb782f2ff618edcae0e39e4; build 7641 (68d8edf2); Apple M2 Metal
Liquid Audio source	19e65845923a7f136442c95137884ec61eb386aa
Qualcomm proxy	QCS8550 (Proxy), Android 12, Hexagon v73, QNN HTP FP16
Comparator catalog	qai-hub-models 0.57.3; QAIRT 2.45.0 scorecards
Full-model golden	PyTorch 2.8.0, Transformers 4.56.1, NVIDIA L4 BF16

References

1. Liquid AI, LFM2.5-Audio model card: <https://huggingface.co/LiquidAI/LFM2.5-Audio-1.5B>
2. Liquid Audio source: <https://github.com/Liquid4AI/liquid-audio>
3. Qualcomm AI Hub model catalog: <https://aihub.qualcomm.com/models/>
4. Qualcomm Zipformer: <https://aihub.qualcomm.com/models/zipformer>
5. Qualcomm Whisper Small: https://aihub.qualcomm.com/models/whisper_small
6. Qualcomm PiperTTS English: https://aihub.qualcomm.com/models/pipertts_en
7. Moonshine Base model card: <https://huggingface.co/UsefulSensors/moonshine-base>
8. Moonshine paper: <https://arxiv.org/abs/2410.15608>
9. Qwen3-0.6B model card: <https://huggingface.co/Qwen/Qwen3-0.6B>
10. Qualcomm Qwen3-0.6B: https://aihub.qualcomm.com/models/qwen3_0_6b
11. Pocket TTS: <https://github.com/kyutai-labs/pocket-tts>
12. Mini-Omni: <https://arxiv.org/abs/2408.16725>
13. LLaMA-Omni2: <https://arxiv.org/abs/2505.02625>